

MOON GUITAR



Moon Guitar is a traditional instrument originating in China. It has a round body with short and slender neck, the sound is distinct and smooth.



Introduction



The Moon Guitar is a 3D modeling and rendering project based on OpenSCAD.

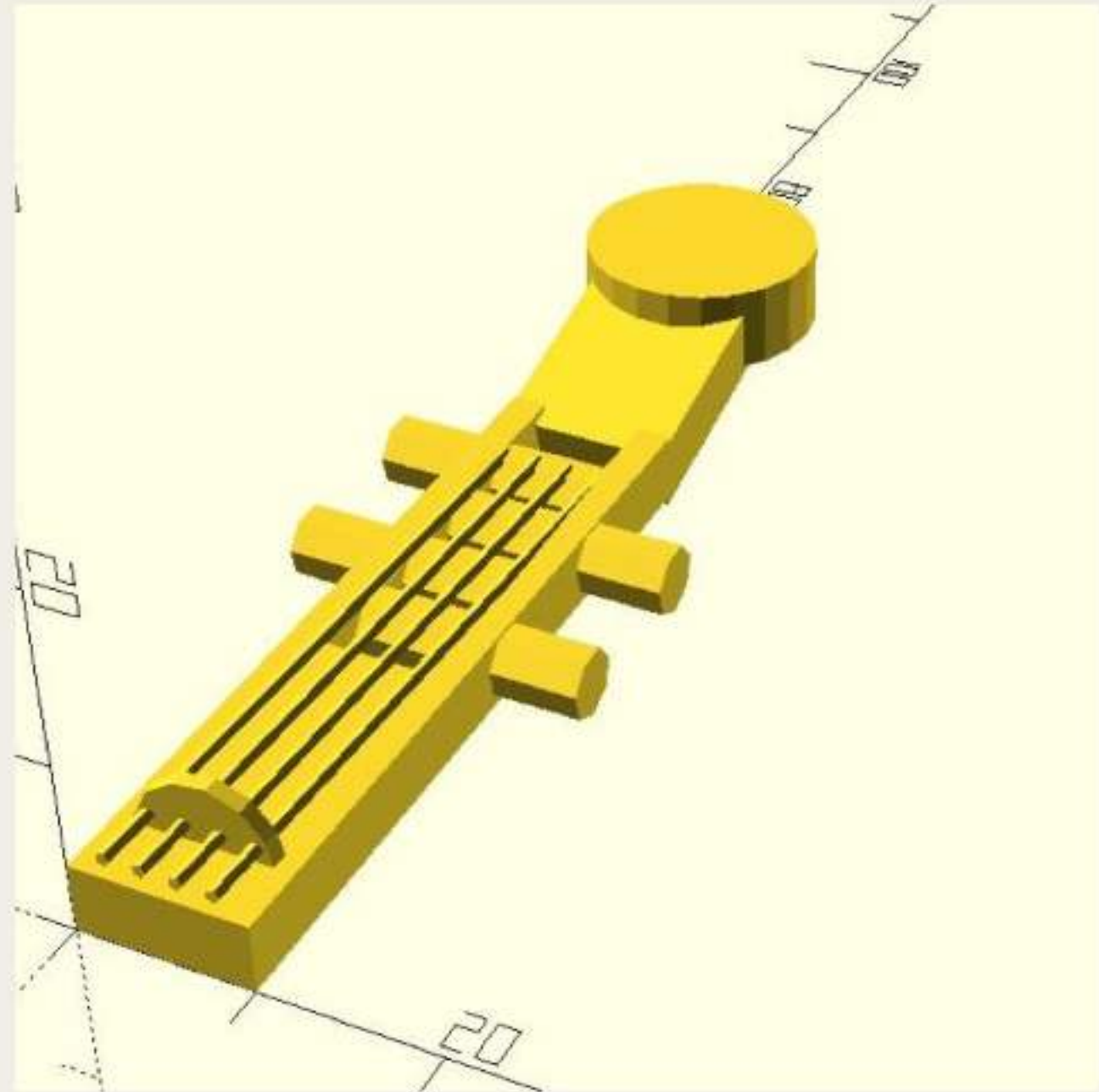
OpenSCAD is an open-source programmable 3D modeling software. I can control the generation of structures through code.

```
function main() {  
  // basic parameters  
  let scale = 5;  
  let radius = 180 / scale; // body size  
  let thickness = 20 / scale;  
  
  // create the bridge body  
  let bridgeRadius = 30 / scale;  
  let bridgeHeight = 15 / scale;  
  
  // create the sound body  
  let halfCylinder = cylinder({r: bridgeRadius, h: bridgeHeight, center: [true, true, false]})  
    .rotateX(90)  
    .intersect(cube({size: [bridgeRadius * 2, bridgeHeight, bridgeHeight * 2], center: [true, true, false]}));  
  
  let bridge1 = halfCylinder.translate([0, -radius + bridgeRadius + 50 / scale + 1, thickness / 2]);  
  
  let bridge2 = bridge1.translate([0, 2 * (radius - bridgeRadius - 50 / scale - 1), 0]);  
  
  // combined together  
  let bridge = union(bridge1, bridge2);  
  
  return bridge;  
}
```


Research

After conducting an initial search for reference images online, it was observed that the soundboards of all Moon Guitar is consist of a circular disc and one or two bridges.

I created these two parts separately and then joined them together and then added some sound holes to the main part



Materials and Rendering

I exported the model and convert it to STL format.

```
facet normal 0 0 -1
outer loop
vertex 7.023251592580619 35.30827009451629 -2
vertex 13.77660356514324 33.25966317040631 -2
vertex 20.000528388705668 29.932906042891627 -2
endloop
endfacet
```

In order to simulate different parts of the Moon Guitar, the model is divided into six parts, and each has unique materials.

```
const material_part2 = new THREE.MeshPhysicalMaterial({
  color: 0xff7744,
  envMap: envTexture,
  metalness: 0.25,
  roughness: 0.5,
  opacity: 1.0,
  transparent: true,
  transmission: 0.99,
  clearcoat: 1.0,
  clearcoatRoughness: 0.25
});
```